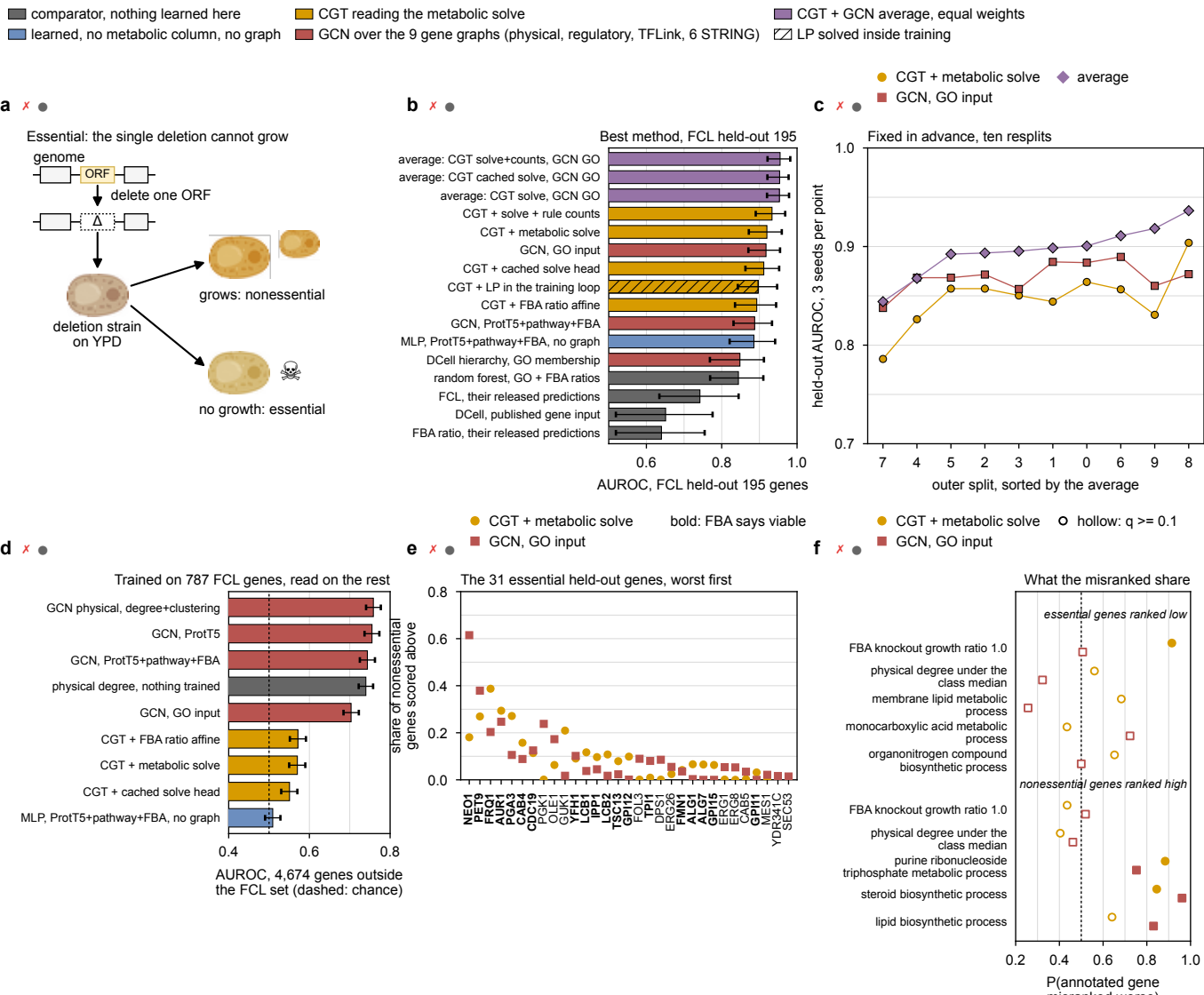

comparator, nothing learned here
learned, no metabolic column, no graph
CGT reading the metabolic solve
GCN over the 9 gene graphs (physical, regulatory, TFLink, 6 STRING)
CGT + GCN average, equal weights
LP solved inside training
a
Essential: the single deletion cannot grow
genome
ORF
delete one ORF
Δ
deletion strain on YPD
grows: nonessential
no growth: essential
b
Best method, FCL held-out 195
average: CGT solve+counts, GCN GO
average: CGT cached solve, GCN GO
average: CGT solve, GCN GO
CGT + solve + rule counts
CGT + metabolic solve
GCN, GO input
CGT + cached solve head
CGT + LP in the training loop
CGT + FBA ratio affine
GCN, ProtT5+pathway+FBA
MLP, ProtT5+pathway+FBA, no graph
DCell hierarchy, GO membership
random forest, GO + FBA ratios
FCL, their released predictions
DCell, published gene input
FBA ratio, their released predictions
0.6
0.8
1.0
AUROC, FCL held-out 195 genes
c
CGT + metabolic solve
average
GCN, GO input
Fixed in advance, ten resplits
held-out AUROC, 3 seeds per point
1.0
0.9
0.8
0.7
7 4 5 2 3 1 0 6 9 8
outer split, sorted by the average
d
Trained on 787 FCL genes, read on the rest
GCN physical, degree+clustering
GCN, ProtT5
GCN, ProtT5+pathway+FBA
physical degree, nothing trained
GCN, GO input
CGT + FBA ratio affine
CGT + metabolic solve
CGT + cached solve head
MLP, ProtT5+pathway+FBA, no graph
0.4
0.6
0.8
AUROC, 4,674 genes outside the FCL set (dashed: chance)
e
CGT + metabolic solve
GCN, GO input
bold: FBA says viable
The 31 essential held-out genes, worst first
share of nonessential genes scored above
0.8
0.6
0.4
0.2
0.0
NEO1 PET9 FRQ1 AUR1 CAB3 CAB4 CDC19 PGK1 OLE1 GUK1 YFH1 LPD1 IPP1 LCB2 TSC13 GPI12 FOL3 TPI1 DPS1 ERG26 FMN1 ALG1 ALG7 GPI15 ERG1 CAB5 GPI11 MES1 YDR341C SEC53
f
CGT + metabolic solve
GCN, GO input
hollow: q >= 0.1
What the misranked share
essential genes ranked low
FBA knockout growth ratio 1.0
physical degree under the class median
membrane lipid metabolic process
monocarboxylic acid metabolic process
organonitrogen compound biosynthetic process
nonessential genes ranked high
FBA knockout growth ratio 1.0
physical degree under the class median
purine ribonucleoside triphosphate metabolic process
steroid biosynthetic process
lipid biosynthetic process
0.2
0.4
0.6
0.8
1.0
P(annotated gene